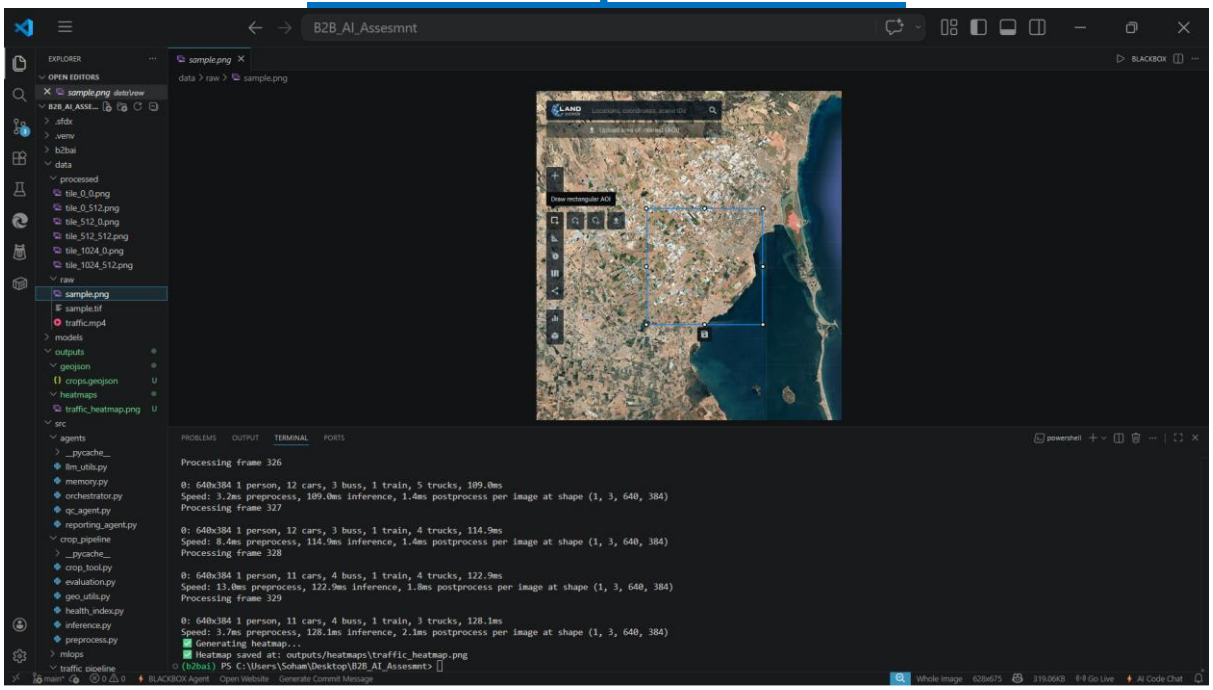
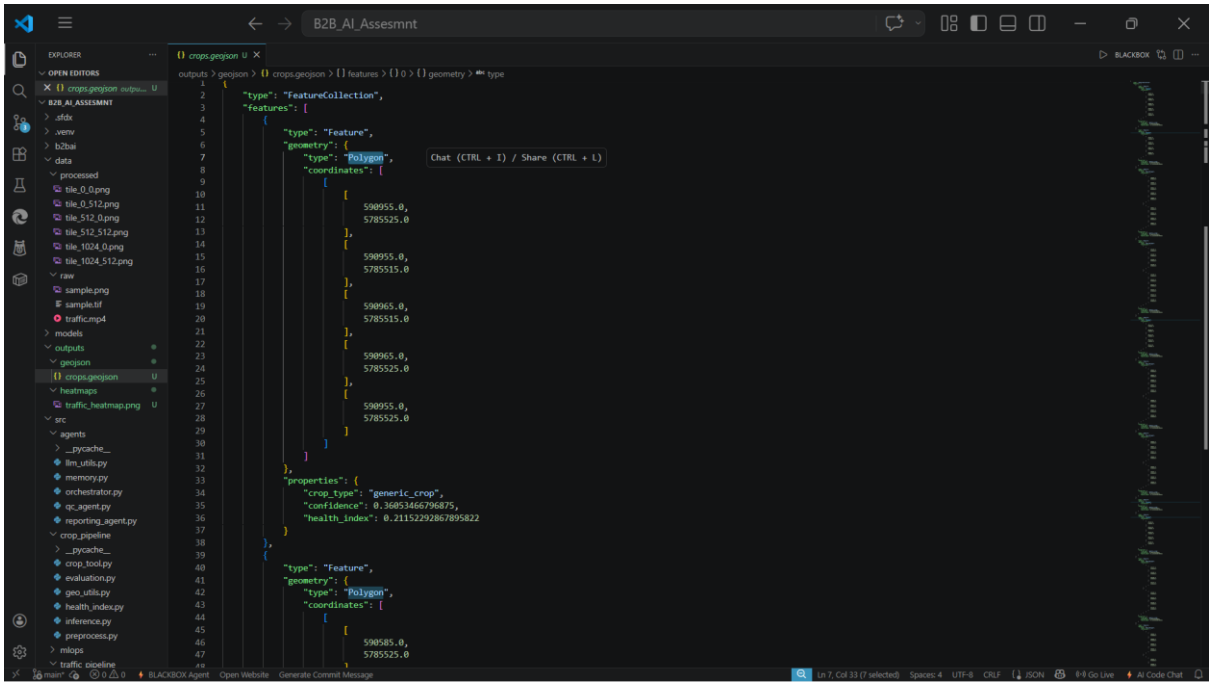
Name: Soham Mali **Task 1 to 4: (Output Images)**

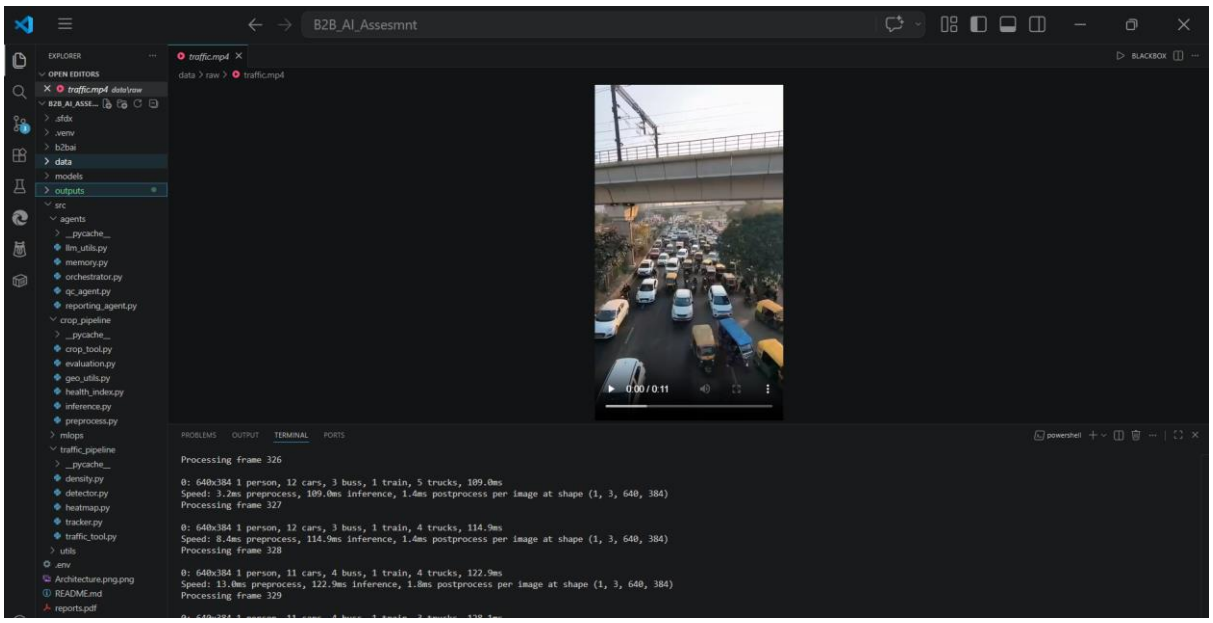
Date: 31.03.2026

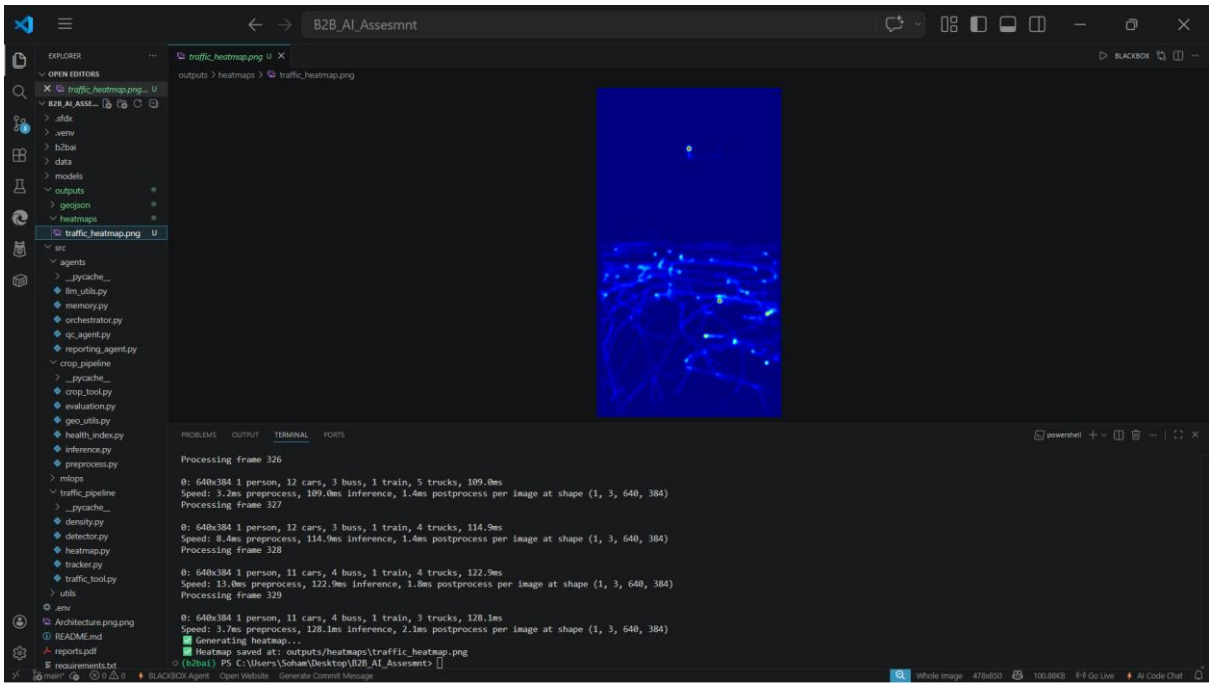
Task 1: Crop Detection:



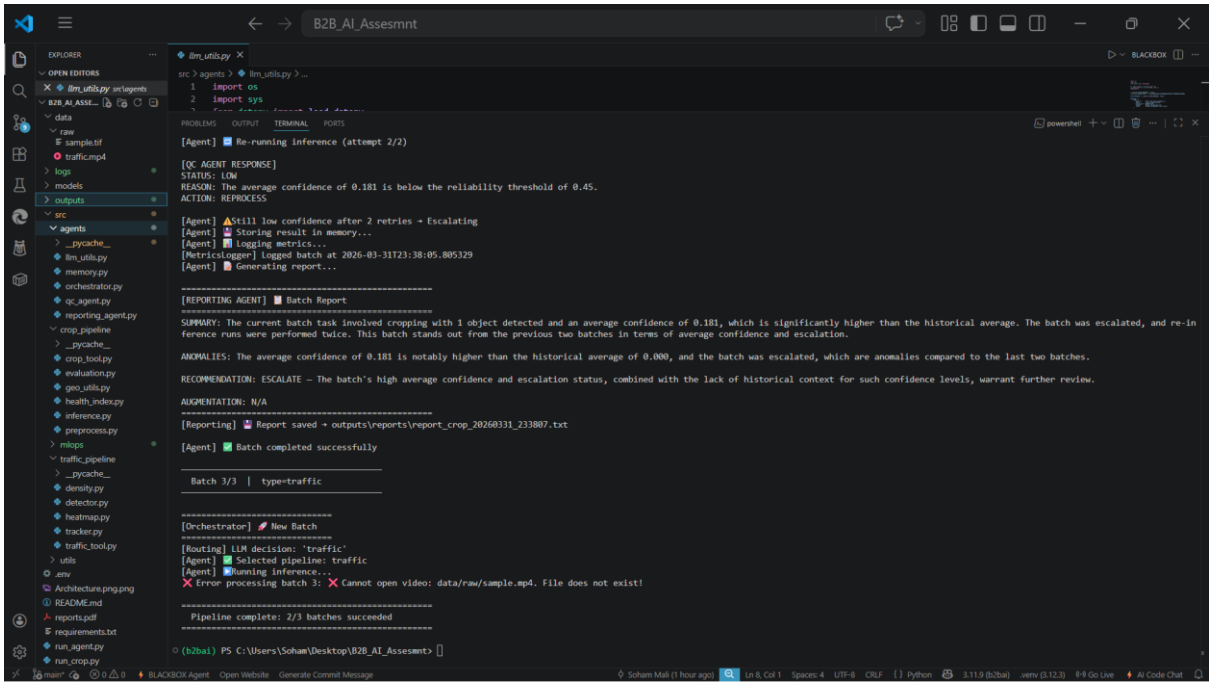


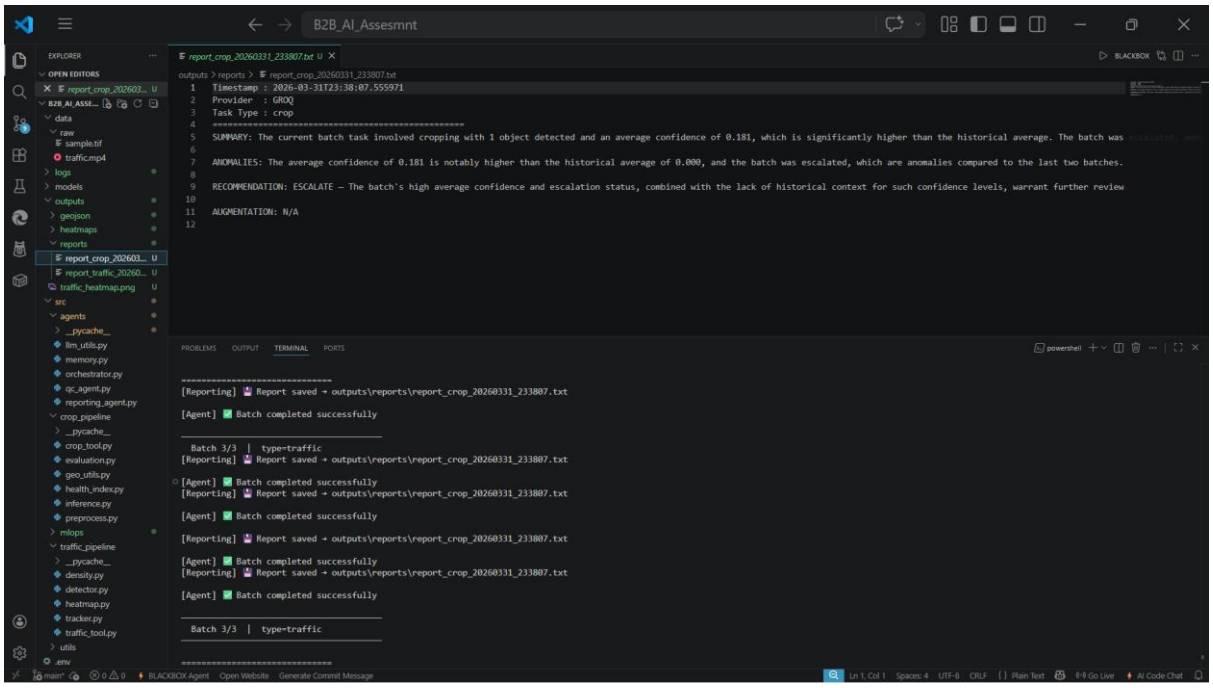
Task 2: Aerial Traffic Detection:





Task 3: Agentic AI Pipeline using LLM and Orchestration Layer





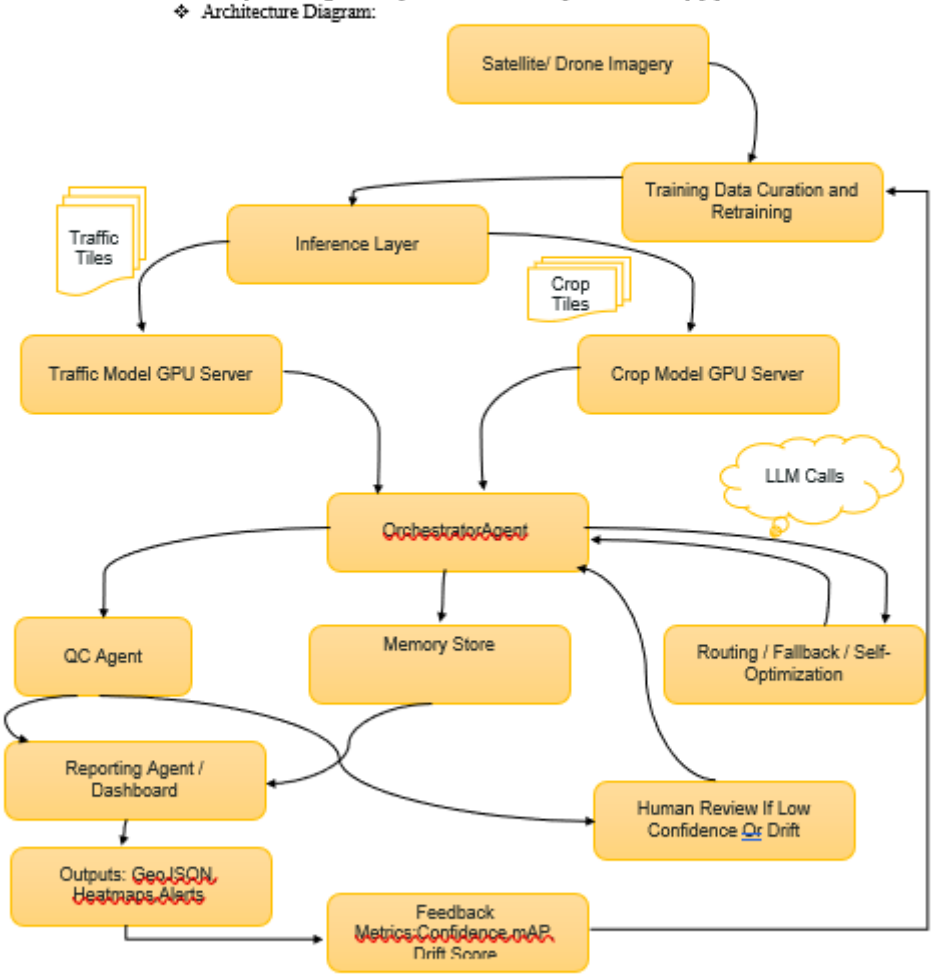
Task 4: Architecture Diagram

Name: Soham Mali
Dt. 31.03.2026

Task 4 — System Integration & Self-Optimization Architecture

4a — Architecture Diagram (8 Marks)

- The system integrates all previous tasks into a production ready pipeline.
- Architecture Diagram:



MINGW64:/c/Users/Soham/Desktop/B2B_AI_Assesmnt

```
 create mode 100644 src/agents/__pycache__/memory.cpython-311.pyc
 create mode 100644 src/agents/__pycache__/orchestrator.cpython-311.pyc
 create mode 100644 src/agents/__pycache__/qc_agent.cpython-311.pyc
 create mode 100644 src/agents/__pycache__/reporting_agent.cpython-311.pyc
 create mode 100644 src/agents/llm_utils.py
 create mode 100644 src/agents/memory.py
 create mode 100644 src/agents/orchestrator.py
 create mode 100644 src/agents/qc_agent.py
 create mode 100644 src/agents/reporting_agent.py
 create mode 100644 src/crop_pipeline/__pycache__/crop_tool.cpython-311.pyc
 create mode 100644 src/crop_pipeline/__pycache__/evaluation.cpython-311.pyc
 create mode 100644 src/crop_pipeline/__pycache__/geo_utils.cpython-311.pyc
 create mode 100644 src/crop_pipeline/__pycache__/health_index.cpython-311.pyc
 create mode 100644 src/crop_pipeline/__pycache__/inference.cpython-311.pyc
 create mode 100644 src/crop_pipeline/__pycache__/preprocess.cpython-311.pyc
 create mode 100644 src/crop_pipeline/crop_tool.py
 create mode 100644 src/crop_pipeline/evaluation.py
 create mode 100644 src/crop_pipeline/geo_utils.py
 create mode 100644 src/crop_pipeline/health_index.py
 create mode 100644 src/crop_pipeline/inference.py
 create mode 100644 src/crop_pipeline/preprocess.py
 create mode 100644 src/mlops/__pycache__/drift.cpython-311.pyc
 create mode 100644 src/mlops/drift.py
 create mode 100644 src/mlops/metrics_logger.py
 create mode 100644 src/traffic_pipeline/__pycache__/density.cpython-311.pyc
 create mode 100644 src/traffic_pipeline/__pycache__/detector.cpython-311.pyc
 create mode 100644 src/traffic_pipeline/__pycache__/heatmap.cpython-311.pyc
 create mode 100644 src/traffic_pipeline/__pycache__/tracker.cpython-311.pyc
 create mode 100644 src/traffic_pipeline/__pycache__/traffic_tool.cpython-311.pyc
 create mode 100644 src/traffic_pipeline/density.py
 create mode 100644 src/traffic_pipeline/detector.py
 create mode 100644 src/traffic_pipeline/heatmap.py
 create mode 100644 src/traffic_pipeline/tracker.py
 create mode 100644 src/traffic_pipeline/traffic_tool.py
 create mode 100644 yolov8n.pt

Soham@DESKTOP-MAVHBII MINGW64 ~/Desktop/B2B_AI_Assesmnt (master)
$ git branch -M main
git push -u origin main
Enumerating objects: 56472, done.
Counting objects: 100% (56472/56472), done.
Delta compression using up to 8 threads
Compressing objects: 100% (44083/44083), done.
error: RPC failed; curl 55 Send failure: Connection was resets
send-pack: unexpected disconnect while reading sideband packet
Writing objects: 100% (56472/56472), 554.75 MiB | 2.06 MiB/s, done.
Total 56472 (delta 11881), reused 56472 (delta 11881), pack-reused 0 (from 0)
fatal: the remote end hung up unexpectedly
Everything up-to-date

Soham@DESKTOP-MAVHBII MINGW64 ~/Desktop/B2B_AI_Assesmnt (main)
$ |
```